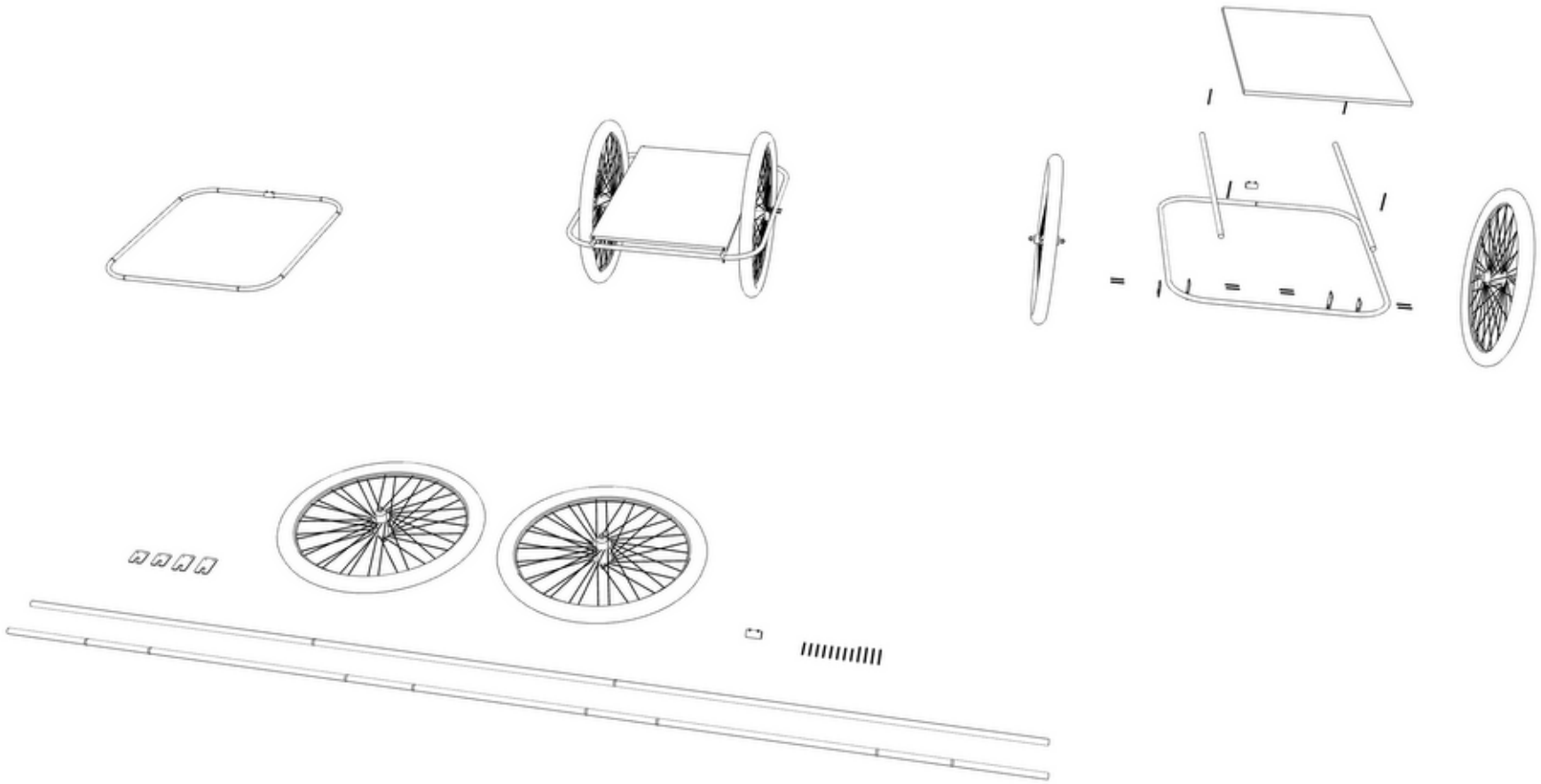
Bike Trailer, version 4

1/2 in EMT conduit, closed loop | two inner bars | 26 in wheels in laser-cut dropouts

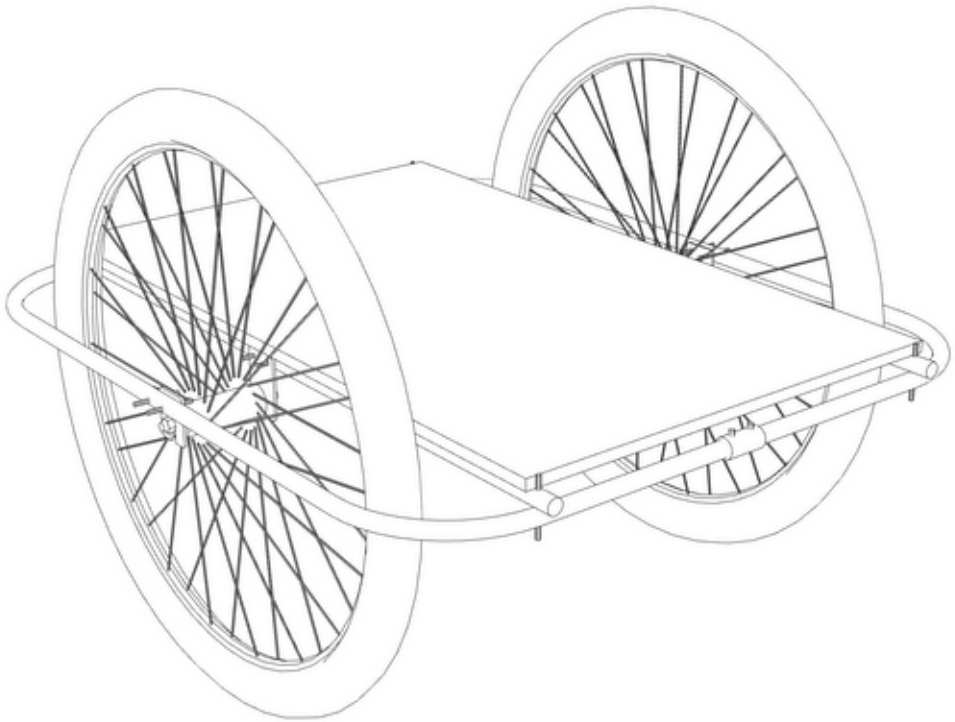


1 Materials > 2 Bent main frame > 3 Assembled trailer > 4 Exploded

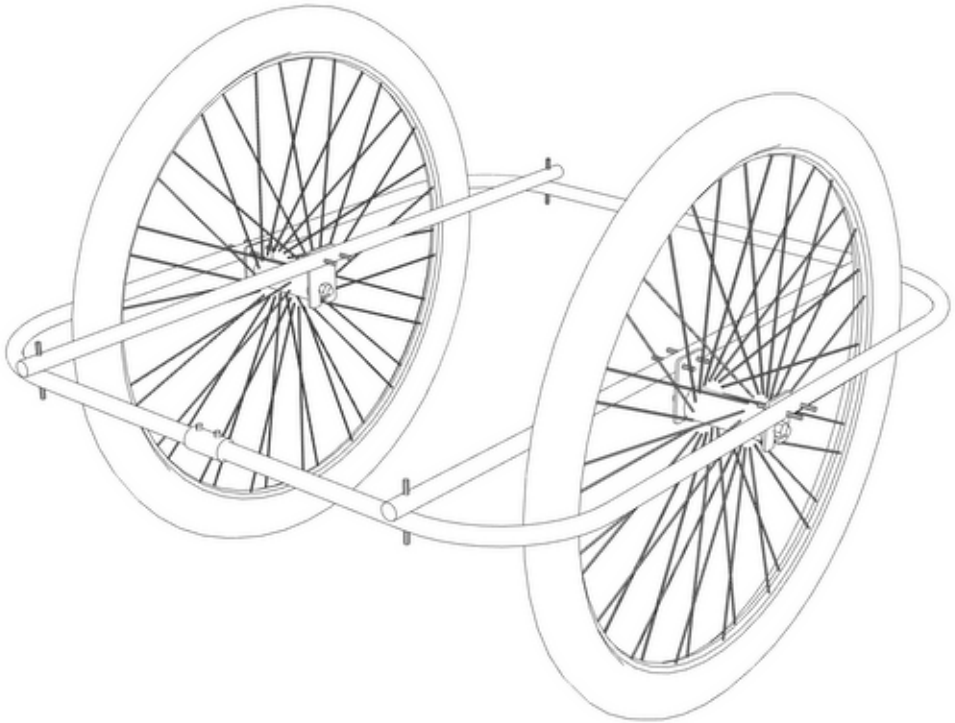
- Main frame: one 10 ft length of 1/2 in EMT bent into a closed loop, 30 7/8 x 34 7/8 in outside. The two ends meet in a set-screw coupling at the front.
- Bends start at the 10, 42, 70 and 102 in marks and use 8 in of conduit each, as in the hand sketch "version 4".
- Inner bars: two 36 in pieces from a second length, 5 1/8 in inside the sides, bolted on top of the front and back.
- Each wheel hangs between a side (short dropout) and an inner bar (long dropout, one pipe higher), so the axle is level.
- Wheel size is estimated from the photo of the dry-fit frame. This version has no hitch (the sketch shows none).

The assembled trailer

With the optional plywood deck, and the bare frame



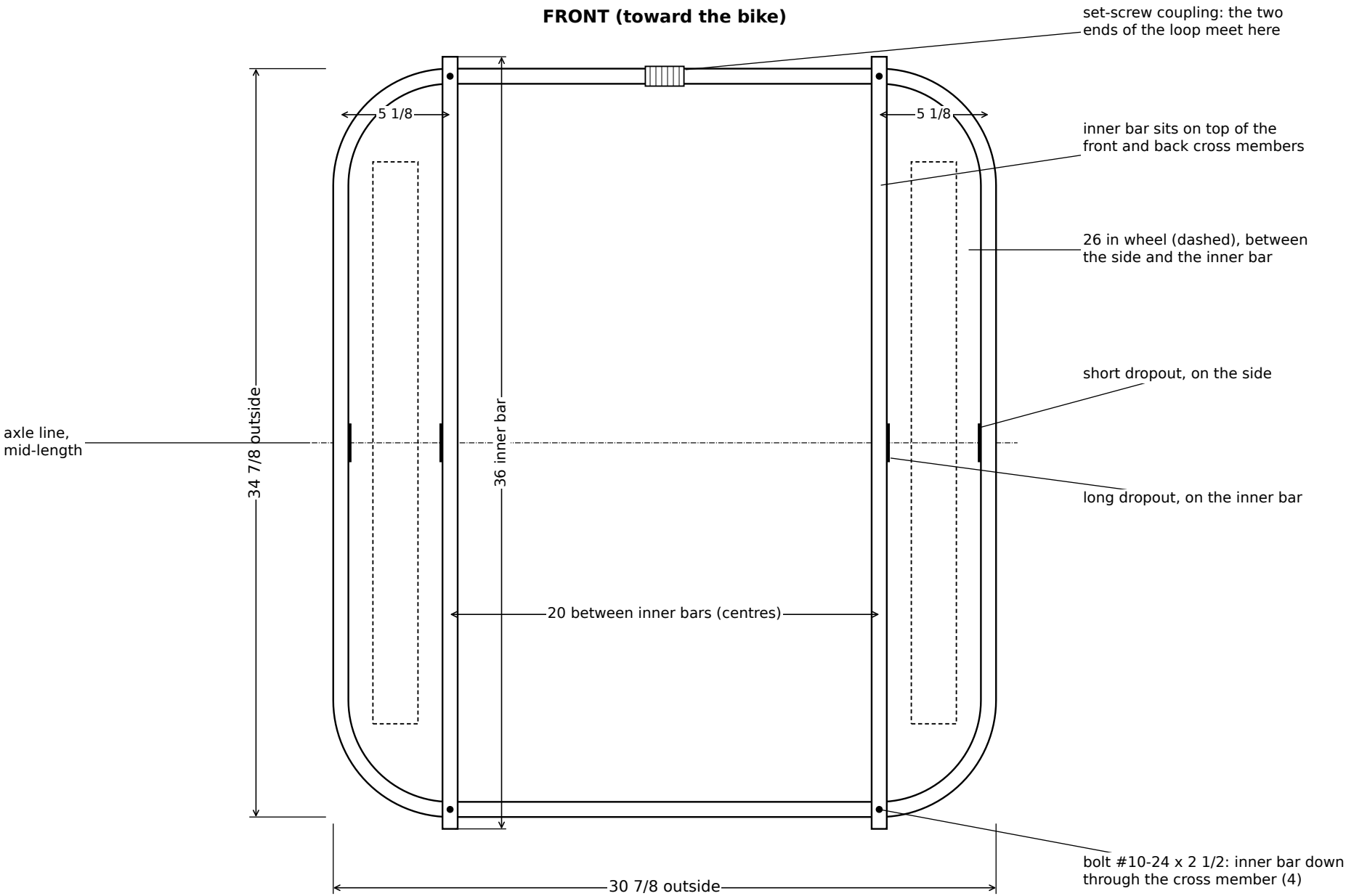
Assembled, with the deck



Frame without the deck

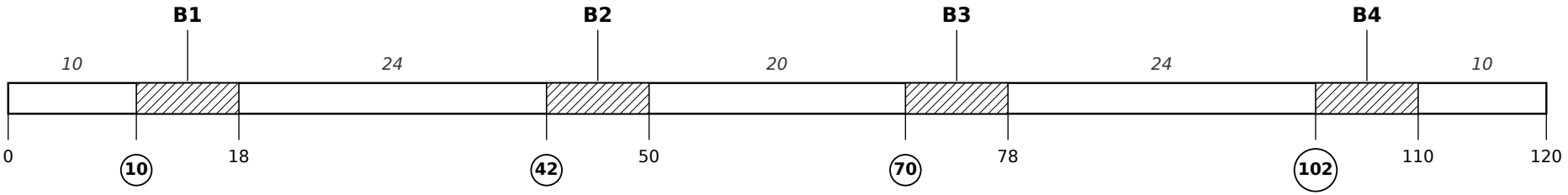
Frame, from above (1:6.5)

Front at the top, as in the sketch. Wheels dashed; dropouts solid black



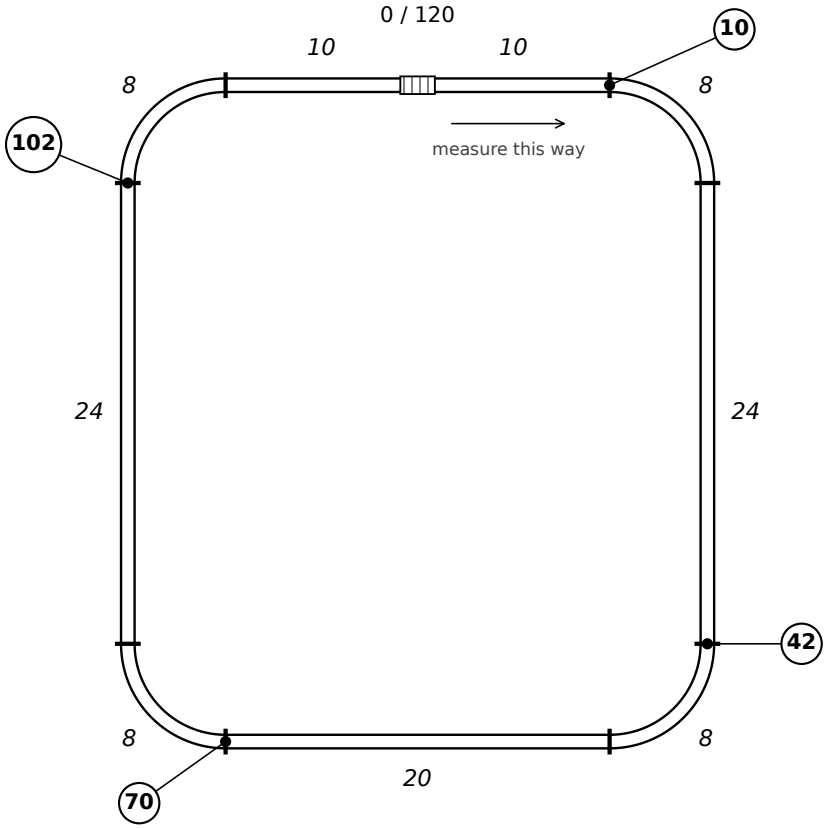
Bending the main frame (one 10 ft conduit)

Marks measured along the pipe from one end. Every bend is 90 degrees, the same way, flat in one plane



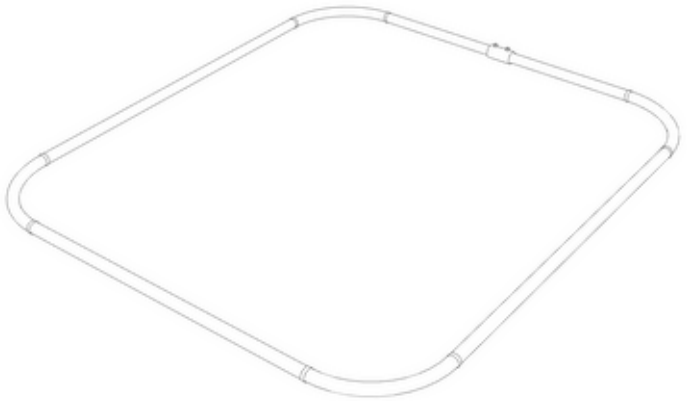
end (joint) — italic numbers: straight lengths; hatched: 8 in of pipe used by each bend; circled: where each bend starts — end (joint)

Where the marks end up (1:10)



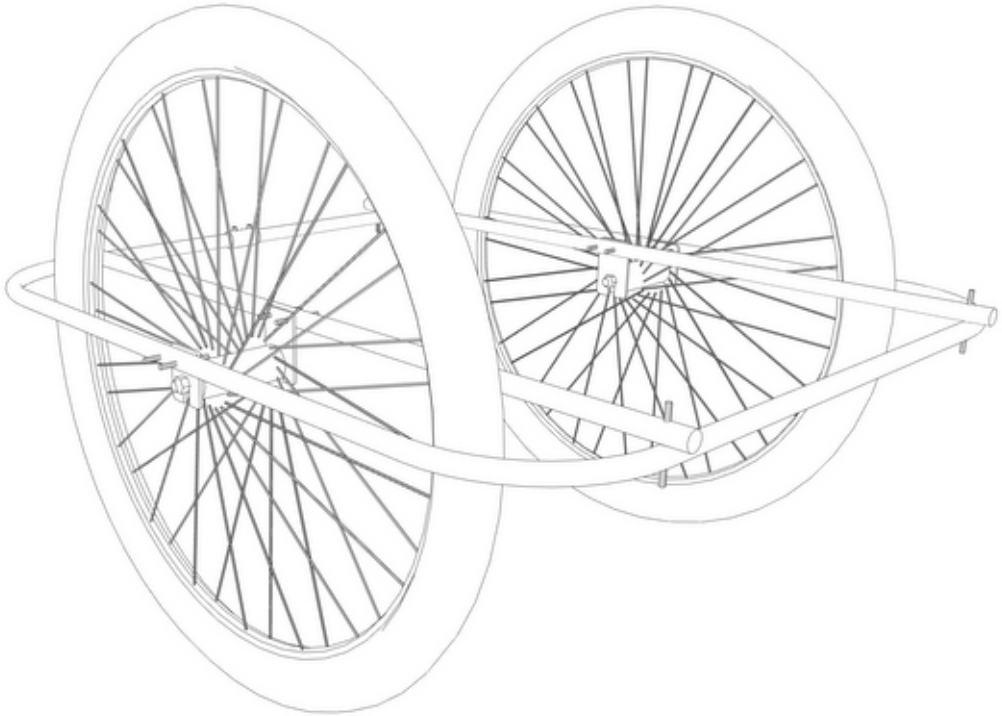
Bend	Start (in)	End (in)	Angle	Corner
B1	10	18	90 deg	front right
B2	42	50	90 deg	back right
B3	70	78	90 deg	back left
B4	102	110	90 deg	front left

Each bend uses 8 in of pipe, a centreline radius of about 5 1/8 in. If your bender uses less, keep the straights at 10, 24, 20, 24 and 10 in and trim what is left over.

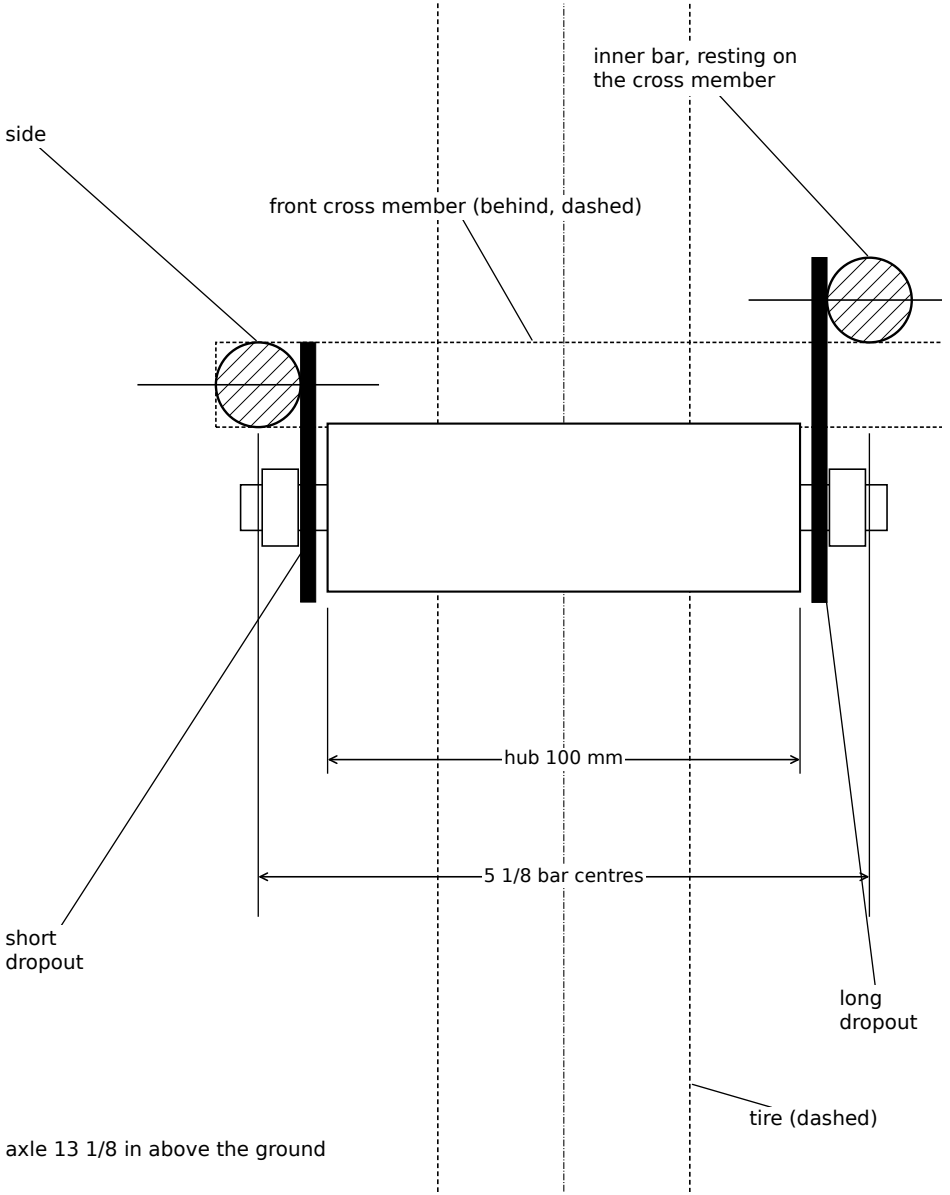


The wheel mount

Short dropout on the side, long dropout on the inner bar, axle level



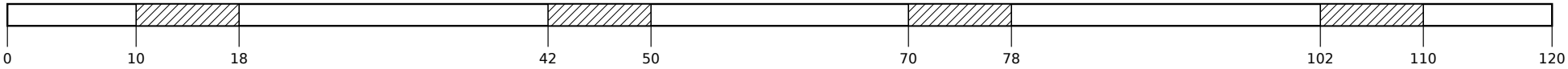
Section through the left axle, looking forward (1:1.6)



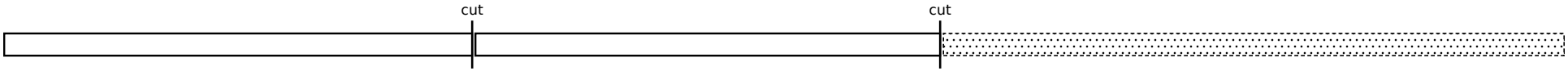
All the pieces, laid out flat (1:12)

Letters match the parts list. Pipe width exaggerated; small parts in the box at 1:2.5

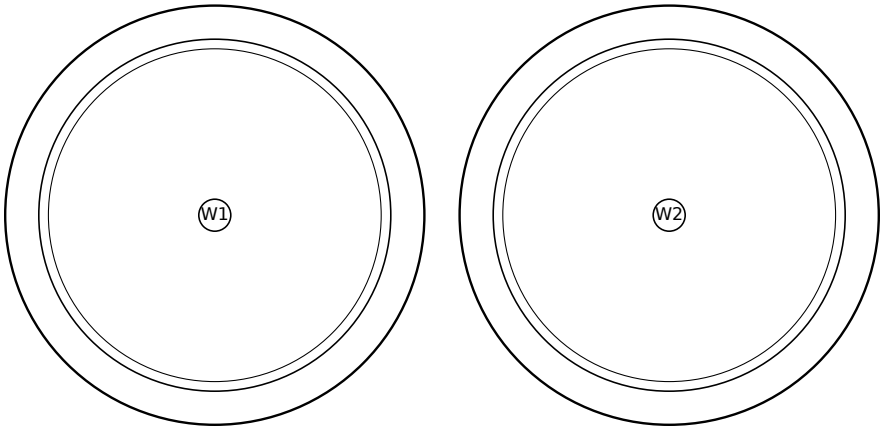
A Main frame: one 10 ft conduit, bend zones hatched



B Inner bars: cut a second 10 ft conduit

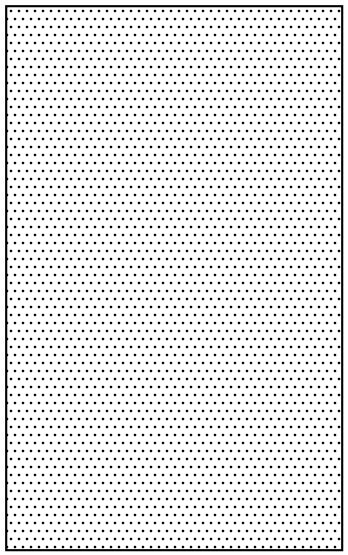


W Wheels (2)



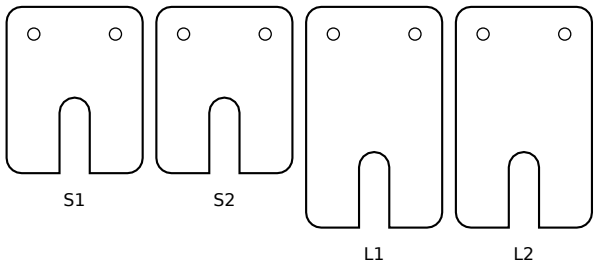
26 x 2.1 in tires, 100 mm front hubs
(size estimated from the photo)

D Deck (optional)

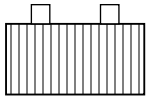


1/2 in plywood
21 x 34 in

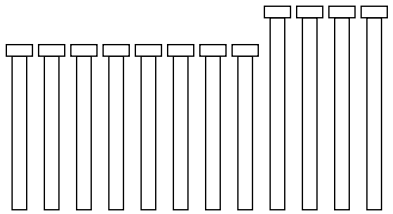
Small parts (1:2.5)



S short dropout (2) L long dropout (2)



C coupling



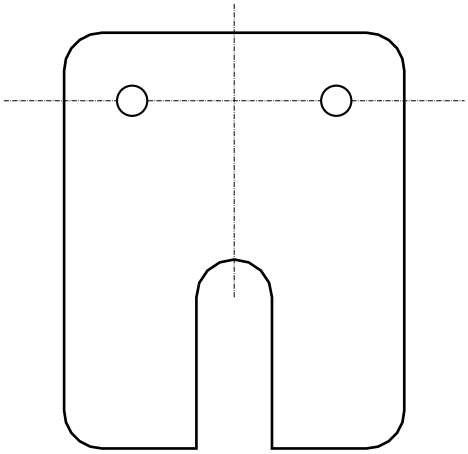
bolts #10-24: 8 x 2 in, 4 x 2 1/2 in

Full-size dropout templates are on the next page.

Dropouts, full size (1:1)

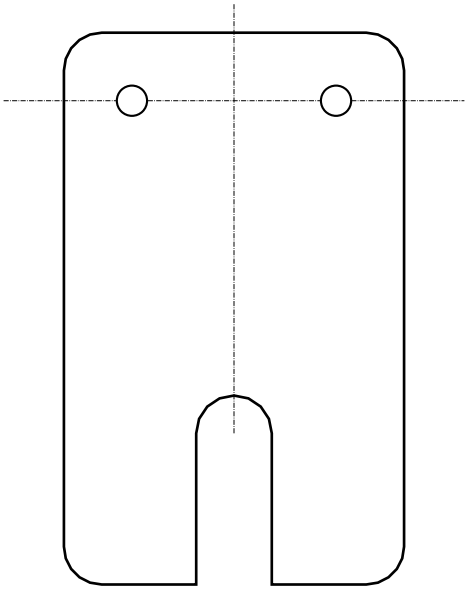
Print at 100% (no scaling). 3.2 mm (0.125 in) stainless; check the squares below

Short dropout (side), make 2

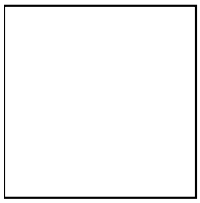


45 x 55 mm (1 3/4 x 2 3/16 in)
2 holes 4 mm (#10 bolts), 27 mm apart
axle slot 10 mm wide

Long dropout (inner bar), make 2

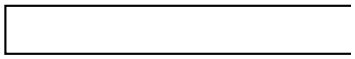


45 x 73 mm (1 3/4 x 2 7/8 in)
2 holes 4 mm (#10 bolts), 27 mm apart
axle slot 10 mm wide



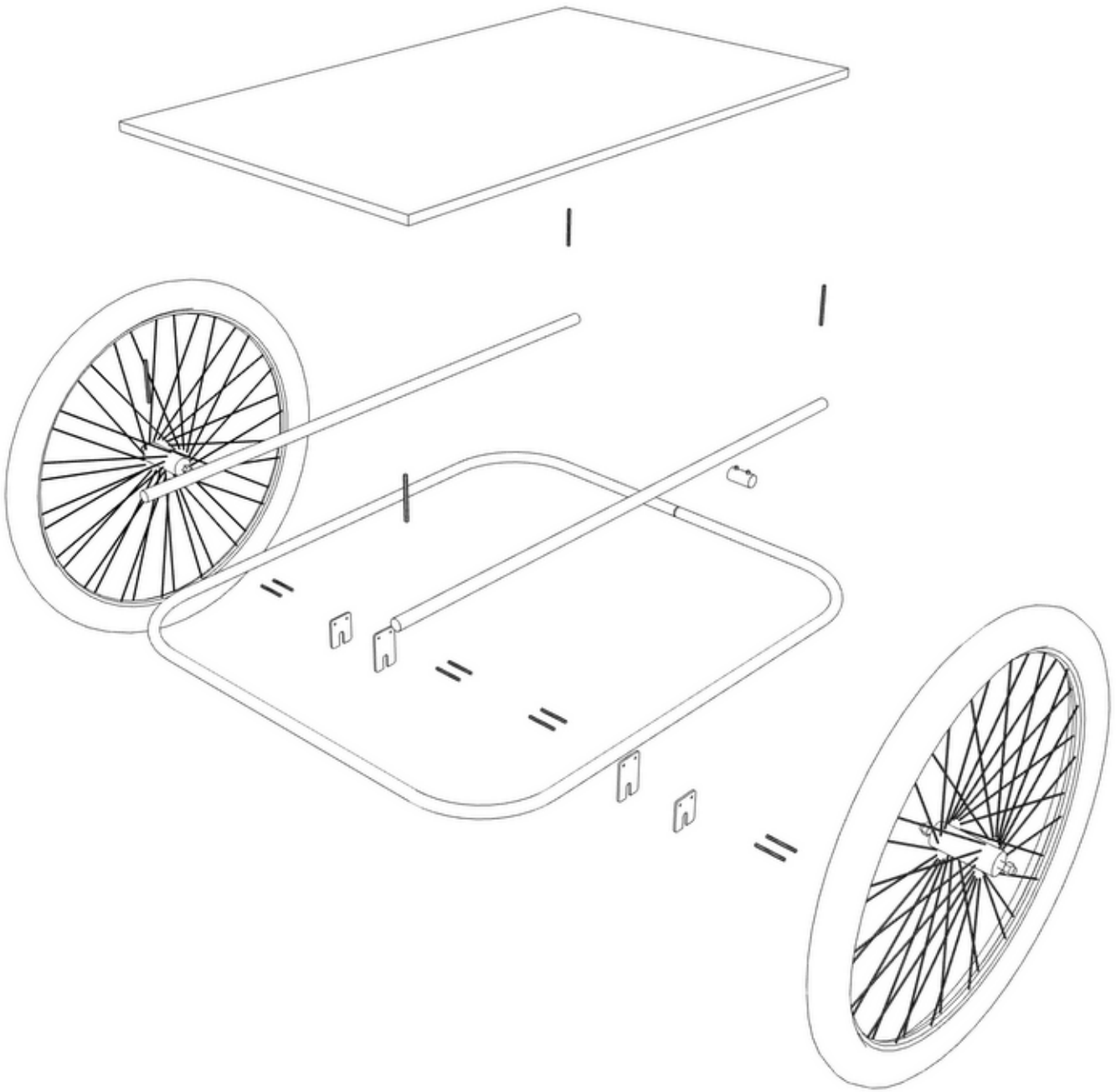
1 in square

50 mm



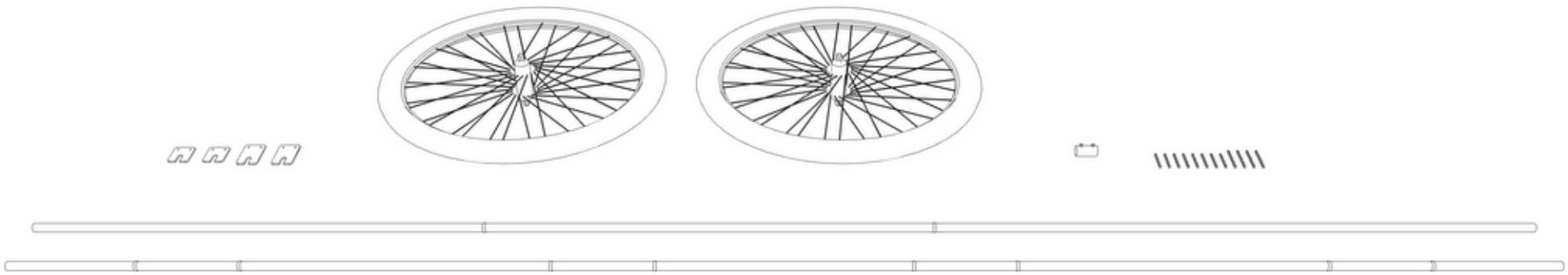
Every part, exploded

Loop and coupling, inner bars and their bolts, dropouts, wheels and deck



Parts list

Quantities match the model



Everything laid out before building

	Part	Qty	Notes
A, B	1/2 in EMT conduit, 10 ft	2	A is bent into the loop (all 120 in); B is cut into the two 36 in inner bars, with 48 in spare
C	1/2 in EMT set-screw coupling	1	joins the two ends of the loop at the front
S	Dropout, short (side)	2	laser cut, 0.125 in stainless; same as version 1
L	Dropout, long (inner bar)	2	18 mm longer, because the inner bar sits one pipe higher
	Bolt #10-24 x 2 in + nut	8	dropouts to the bars, 2 each
	Bolt #10-24 x 2 1/2 in + nut	4	inner bars down through the front and back cross members
W	26 in wheel	2	100 mm front hubs; size estimated from the photo
D	Plywood deck, 1/2 in, 21 x 34 in	1	optional
	Hitch	-	not in the sketch; the 48 in spare could make a tongue
	Tools		1/2 in EMT bender, tubing cutter or hacksaw, drill and 11/64 in bit, wrenches